

Sarthak Jain

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EDUCATION

- Carnegie Mellon University** / *B.S. Electrical & Computer Engineering and Robotics* / **GPA 3.8/4.0** Expected May 2028
- **Coursework:** Signals & Systems, Structure & Design of Digital Systems, Introduction to Computer Systems, Embedded Systems, Computer Vision, Fundamentals of Controls, Electronics & Circuits, Principles of Imperative Computation, Linear Algebra
 - **Teaching Assistant, 18-290 Signals and Systems (Fall 2026):** Supported **100+ students**, by teaching recitations and office hours on Fourier/Laplace transforms, FFT, sampling, and filter design, and by writing and grading problem sets and exams.

EXPERIENCE

- AirLab, CMU Robotics Institute** / *Robotics Autonomy Researcher* May 2026 – Present
- Enabled a drone to sense nearby walls with **zero added sensors** (no camera or LiDAR), reaching **under 3 cm** distance and **under 4°** angle error, by independently prototyping and evaluating a **physics-informed machine learning** model in **PyTorch** that processes the drone's own thrust, RPM, and IMU **sensor signals**.
 - Tracked distance continuously from noisy time-series data, by designing a temporal-encoder **state-estimation algorithm** that runs recursive **inference** in real time.
 - Produced a **120,000-sample** labeled training set without risking hardware, by applying **numerical methods** in an **ANSYS Fluent CFD** digital twin of near-surface airflow and closing the sim-to-real gap with domain randomization.
- ForAll Lab, Carnegie Mellon** / *Neuroscience / Signals Engineer* Summer 2026
- Aligned brain-computer-interface recordings from separate machines to **within 2 ms**, a **~30×** reduction in timing spread, by developing a **signal-processing algorithm** that models clock drift with **least-squares regression** in **NumPy/SciPy**.
 - Recovered **100% of event markers** with **0 mismatched triggers** in validated sessions, by de-clustering noisy trigger bursts and building an automated evaluation pipeline that flags dropped triggers and quarantines out-of-tolerance data.
- Kantor Lab, Carnegie Mellon** / *Research Assistant* 2025 – Present
- Helped **win the Farm Robotics Challenge (\$25K)**, by co-training and deploying real-time **YOLOv8 (PyTorch)** **computer-vision inference** that detects and grades apple-tree disease on an autonomous robot's onboard compute.
 - Closed the loop from perception to actuation, by building the **ROS 2** (Python/C++) pipeline that turns detections into robotic-arm motion, and **profiling** and fixing **latency** and synchronization bottlenecks between vision and control.

PROJECTS

- Edge-AI Health Vest** | *TinyML, INT8 quantization, sensor fusion, Kalman filtering* 2024 – 2025
- Achieved **under 8 ms** inference on a **sub-milliwatt** power budget with no phone or cloud, by **pruning and INT8-quantizing** the health model to run on the vest's microcontroller.
 - Cut motion noise **43%** on a sprinting athlete, by fusing **7 biosignal channels** (heart rate, SpO2, HRV, temperature, IMU) with a denoising autoencoder and adaptive **Kalman filter**.
 - Put the vest in the hands of **500+ blind football players**, by partnering with Scout Me to produce and distribute it.
- Pipelined RISC-V Processor** | *SystemVerilog, computer architecture, Vivado, QuestaSim* 2025
- Kept **5 instructions in flight** at once with **0 hazard failures**, by designing hazard-detection and forwarding logic for an RV32I CPU in SystemVerilog.
 - Verified **100% cycle-accurate** behavior, by building a QuestaSim testbench and pipeline visualizer; synthesized in Vivado.
- Shadow Fight** | *C++17, QNX RTOS, OpenCV, dual Raspberry Pi, real-time systems* 2025
- Built a game you fight with your own shadow, with **under 80 ms** camera-to-screen latency, by writing a **C++17** vision and rendering system split across two Raspberry Pis on the **QNX real-time OS**.
 - Kept traffic **under 12 kbps** at 30 fps, by designing a **~50-byte UDP protocol** with timestamp-matched buffering.
- AI Air Hockey** | *Computer vision, real-time control, OpenCV, embedded C* 2025
- Blocked human shots in **under 50 ms**, by tracking the puck in OpenCV, predicting wall bounces, and profiling the camera-to-motor loop.

TECHNICAL SKILLS

AI Systems & Inference Optimization: AI inference, INT8 quantization, pruning, TinyML, edge inference, TensorRT, kernel-level optimization, latency & power profiling, benchmarking, PyTorch, YOLOv8, physics-informed ML, sim-to-real

Signal Processing & Numerical Methods: Fourier & Laplace transforms, DFT/FFT, sampling, filter design, spectral analysis, sensor fusion, Kalman filtering, state estimation, robust regression, time-series analysis, MATLAB

Computer Architecture & Digital Design: SystemVerilog, RTL design, pipelining, hazard detection & forwarding, testbench verification, RISC-V, FPGA, Vivado, QuestaSim

Low-Level & Systems Programming: C/C++, Python, Rust, embedded C firmware, QNX RTOS, Linux, device drivers, I2C/SPI/UART, multithreading, POSIX sockets, CMake, Git, Docker, NumPy/SciPy, OpenCV, ROS 2